

Chapter 3: Force and Pressure

Textbook Exercise Solutions & Answers



1. Write proper word in the blank space

(a) The SI unit of force is the (dyne, newton, joule)

The SI unit of force is the newton.

(b) The air pressure on our body is equal to the pressure. (atmospheric, sea bottom, space)

The air pressure on our body is equal to the atmospheric pressure.

(c) For a given object, the buoyant force in liquids of different is (the same, density, different, area)

For a given object, the buoyant force in liquids of different density is the same.

(d) The SI unit of pressure is (N/m^3 , N/m^2 , kg/m^2 , Pa/m^2)

The SI unit of pressure is N/m^2 .

2. Make a match.

Group 'A'	Group 'B'
1. Fluid	e. Same pressure in all directions
2. Blunt knife	d. Lower pressure
3. Sharp needle	a. Higher pressure
4. Relative density	c. Specific gravity
5. Hectopascal	b. Atmospheric pressure

3. Answer the following questions in brief.

(a) A plastic cube is released in water. Will it sink or come to the surface of water?

It will come to the surface of water.

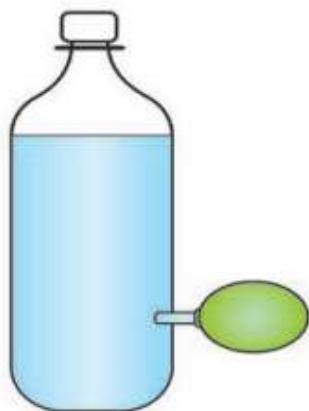
[Note: This is because its density is less than that of water. When it floats, the unbalanced force acting on it is zero.]

(b) Why do the load carrying heavy vehicles have large number of wheels?

The pressure produced by a given force depends on the area of the surface on which the force acts. The greater the surface area, the less is the pressure produced. Load carrying heavy vehicles have large number of wheels so that the load (weight, force) is distributed over large surface area of the wheels in contact with the road. Hence, the pressure decreases and the tyres do not burst.

Activity 1:

Take a plastic bottle. Take a 10 cm long piece of a glass tube on which a rubber balloon can be fitted. Warm up one end of the glass tube and gently push it into the bottle at about balloon inflates. What is observed? The pressure of water acts on the side of the bottle as well.



Pressure of Liquid

[Note: Here, the area of cross section of the tube remains the same. As the level of the water in the bottle rises, the mass of the water increases resulting in increase in the weight. As the applied force increases, the pressure increases. Therefore, the balloon increases in size.]

Activity 2:

Take a plastic bottle. Pierce it with a thick needle (or with a hot nail) at the points 1, 2, 3 as shown in the Fig. Fill water in the bottle up to full height. As shown in the figure, water jets will be seen emerging and projecting out. The water jet emerging from the hole at the top will fall closest to the bottle. The jet from the lowest hole falls farthest from the bottle.

Also, jets coming out from the two holes at the same level fall at the same distance from the bottle. What is understood from this? At any one level, the liquid pressure is the same. Also, the pressure increases as the depth of the liquid increases.



Level and Pressure of Liquid

(c) How much pressure do we carry on our heads? Why don't we feel it?

The air pressure at the sea level is about 101×10^3 Pa. This is the pressure that we carry on our heads. The cavities in our body are filled with air, and arteries and veins are filled with blood. Their pressure balances the pressure due to the atmosphere. Hence, we don't feel the atmospheric pressure.

4. Why does it happen?

(a) Why does it happen? A ship dips to a larger depth in freshwater as compared to marine water.

The buoyant force acting on a body is proportional to the density of the fluid in which the body is immersed. The density of freshwater is less than that of marine water. Hence, the buoyant force on a body in freshwater is less than that in marine water. Therefore, a ship dips to a larger depth in freshwater as compared to marine water.

(b) Why does it happen? Fruits can easily be cut with a sharp knife.

1. It is easy to cut vegetables, fruits with a sharp knife. A blunt knife does not work here.
2. The force exerted perpendicularly on a unit area is called 'pressure'

$$\text{Pressure} = \frac{\text{Force}}{\text{Area on which the force is applied}}$$

3. Presently we are considering only the force acting on an area in a direction perpendicular to it.

(c) Why does it happen? The wall of a dam is broad at its base.

1. The pressure at a point in a liquid is proportional to the height of the liquid column above it. Hence, the pressure of water in a dam is much greater at the bottom of the dam than at the top.
2. To withstand this high pressure, the wall of a dam is made stronger and thicker (broad) at the base than at the top.

(d) Why does it happen? If a stationary bus suddenly speeds up, passengers are thrown in the backward direction.

1. When passengers sit or stand in a stationary bus, they are in a state of rest. When the bus suddenly speeds up, the lower (parts of their body in contact with the bus acquire the speed of the bus.
2. The upper parts of their body, however, continue to be in the state of rest due to inertia. Hence, they are thrown in the backward direction.

5. Complete the following tables:

(a) Complete the table using the formula $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$:

Formula:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Step-by-step Calculation:

1. Row 1: Given Mass = 350 kg, Volume = 175 m³

$$\text{Density} = \frac{350 \text{ kg}}{175 \text{ m}^3} = \mathbf{2 \text{ kg/m}^3}$$

2. Row 2: Given Volume = 190 m³, Density = 4 kg/m³

$$\text{Mass} = \text{Density} \times \text{Volume} = 4 \text{ kg/m}^3 \times 190 \text{ m}^3 = \mathbf{760 \text{ kg}}$$

Mass (kg)	Volume (m ³)	Density (kg/m ³)
350	175	2
760	190	4

(b) Complete the table using the formula $\text{Relative Density} = \frac{\text{Density of metal}}{\text{Density of water}}$:

Formula:

$$\text{Relative Density} = \frac{\text{Density of metal}}{\text{Density of water}}$$

Step-by-step Calculation:

1. Row 1: Given Density of water = 10^3 kg/m^3 , Relative Density = 5

$$\text{Density of metal} = \text{Relative Density} \times \text{Density of water} = 5 \times 10^3 \text{ kg/m}^3 = \mathbf{5 \times 10^3 \text{ kg/m}^3}$$

2. Row 2: Given Density of metal = $8.5 \times 10^3 \text{ kg/m}^3$, Density of water = 10^3 kg/m^3

$$\text{Relative Density} = \frac{8.5 \times 10^3 \text{ kg/m}^3}{10^3 \text{ kg/m}^3} = \mathbf{8.5}$$

Density of Metal (kg/m ³)	Density of Water (kg/m ³)	Relative Density
5×10^3	10^3	5
8.5×10^3	10^3	8.5

(c) Complete the table using the formula $\text{Pressure} = \frac{\text{Weight}}{\text{Area}}$:

Formula:

$$\text{Pressure} = \frac{\text{Weight}}{\text{Area}}$$

Step-by-step Calculation:

1. Row 1: Given Area = 0.04 m^2 , Pressure = $20,000 \text{ N/m}^2$

$$\text{Weight} = \text{Pressure} \times \text{Area} = 20,000 \text{ N/m}^2 \times 0.04 \text{ m}^2 = \mathbf{800 \text{ N}}$$

2. Row 2: Given Weight = 1,500 N, Area = 500 m^2

$$\text{Pressure} = \frac{1,500 \text{ N}}{500 \text{ m}^2} = \mathbf{3 \text{ N/m}^2} \quad (\text{or } 3 \text{ Pa})$$

Weight (N)	Area (m ²)	Pressure (N/m ² or Pa)
800	0.04	20,000
1,500	500	3

6. The density of a metal is $10.8 \times 10^3 \text{ kg/m}^3$. Find the relative density of the metal.

Given:

Density of the metal (ρ_{metal}) = $10.8 \times 10^3 \text{ kg/m}^3$

Density of water (ρ_{water}) = $1.0 \times 10^3 \text{ kg/m}^3$

Formula:

$$\text{Relative density of metal} = \frac{\text{Density of metal}}{\text{Density of water}}$$

Calculation:

$$\text{Relative density} = \frac{10.8 \times 10^3 \text{ kg/m}^3}{10^3 \text{ kg/m}^3} = \mathbf{10.8}$$

Answer:

The relative density of the metal is **10.8** (it is a ratio of identical units, hence it is dimensionless).

7. The volume of an object is 20 cm^3 and the mass is 50 g . The density of water is 1 gm-^3 . Will the object float on water or sink in water?

Given:

Mass of the object (m) = 50 g

Volume of the object (V) = 20 cm^3

Density of water (ρ_{water}) = 1 g/cm^3

Formula:

$$\text{Density of the object } (\rho) = \frac{\text{Mass}}{\text{Volume}} = \frac{m}{V}$$

Calculation:

$$\rho = \frac{50 \text{ g}}{20 \text{ cm}^3} = \mathbf{2.5 \text{ g/cm}^3}$$

Conclusion:

The density of the object (2.5 g/cm^3) is greater than the density of water (1.0 g/cm^3). Since the object is denser than water, the downward gravitational force is greater than the upward buoyant force.

Answer: Therefore, the object will **sink in water**.

8. The volume of a plastic-covered sealed box is 350 cm^3 and the box has a mass 500 g . Will

the box float on water or sink in water? What will be the mass of water displaced by the box?

Given:

Mass of the sealed box (m) = 500 g

Volume of the sealed box (V) = 350 cm³

Density of water (ρ_{water}) = 1 g/cm³

Part 1: Will the box float or sink?

$$\text{Density of the box } (\rho) = \frac{\text{Mass}}{\text{Volume}} = \frac{500 \text{ g}}{350 \text{ cm}^3} \approx \mathbf{1.43 \text{ g/cm}^3}$$

Since the density of the box (1.43 g/cm³) is greater than the density of water (1.0 g/cm³), the box will **sink in water**.

Part 2: Mass of water displaced by the box:

As the box completely sinks, the volume of water displaced is equal to the volume of the box:

$$V_{\text{water displaced}} = 350 \text{ cm}^3$$

Using the relation Mass = Volume $\times$ Density:

$$\text{Mass of water displaced} = V_{\text{water displaced}} \times \rho_{\text{water}} = 350 \text{ cm}^3 \times 1 \text{ g/cm}^3 = \mathbf{350 \text{ g}}$$

Answer:

1. The box will **sink in water** (since its density 1.43 g/cm³ > 1 g/cm³).
2. The mass of water displaced by the box is **350 g**.